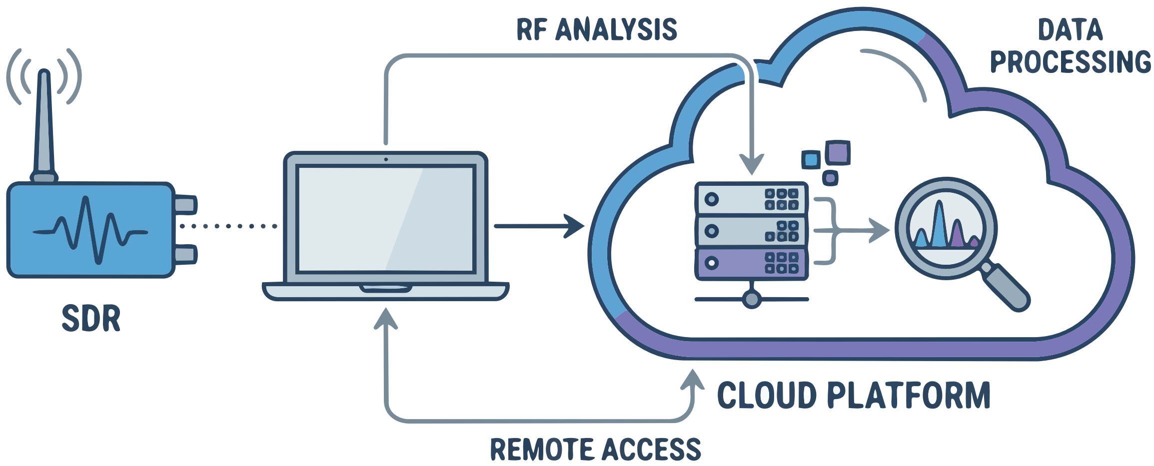
Software Defined Radio and Radio Frequency Analysis on the Cloud: An Overview

Software defined radio (SDR) and radio frequency analysis on cloud-based platforms are transforming wireless communication.



Software defined radio (SDR) and spectrum analysis are powerful technologies in the domains of wireless communication and signal intelligence. SDR is used by the military, defence services, aviation industry, intelligence services, in corporate communications, as well as in other domains that need privacy, security and integrity. Traditional radio systems were dependent on fixed hardware devices and components for filtering, modulation, demodulation and signal intelligence. SDR shifts most of these functions to software, enabling dynamic configuration and reprogramming. The increasing demand for wireless connectivity, satellite communications, IoT networks and spectrum monitoring has drastically accelerated the adoption of SDR.

SDR enables components that were traditionally implemented in hardware to be deployed and controlled using software on embedded systems, microcontroller-based systems, personal computers or cloud infrastructure. SDR devices receive and transmit radio signals over a wide spectrum and

Key use cases and application domains of SDR

- Radio frequency and spectrum monitoring
- Signal intelligence
- Communications intelligence
- Electronic intelligence
- Military intelligence
- Satellite reception
- ADS-B aircraft tracking
- Aircraft crash data analysis
- Aircraft accident cause investigation
- Weather satellite decoding
- IoT protocol analysis
- Amateur radio experimentation
- GSM research testing
- 5G waveform prototyping
- LTE network analysis
- Cognitive radio research
- RF security testing
- Interference detection
- Disaster communication systems
- Remote sensing studies
- Spectrum occupancy mapping
- Military communication research
- Digital modulation experiments
- Wireless protocol development
- V2V communication testing